

CS383 - Software Engineering

Software Design (I)

System modeling

- Create models of intended system and its subsystems from different point of views
 - Models lack of implementation details
 - UML graphical notation is used to create models
- For existing systems, models can help clarify what are system functionalities
 - Help identify strengths and weaknesses
- For new systems, models can help during requirement engineering process
 - Help explain clearly proposed requirements to clients/customers
 - Help to design abstract system, its components, system proposals and documentation (e.g. for implementation)
- For model-driven engineering
 - Models are used to do partial, or full implementation of target systems

System modeling (cont.)

- There are four main perspectives to which models are created:
 1. Context or environment perspective: system boundaries and external systems
 2. Interaction perspective: modeling interactions between system components and the whole system with surrounding environment
 3. Structural perspective: modeling the organisation of the system and its internal elements (as components) and data structure of the system
 4. Behavioral perspective: model the activity of the system and how it respond to events

UML - Unified Modeling Language

- A language of a set of 13 different diagram types that may be used to model software systems.
- It emerged from work in the 1990s on object-oriented modeling where similar object-oriented notations were integrated to create the UML.
- A major revision (UML2) was finalized in 2004.
- The UML is universally accepted as the standard approach for developing models
- One of best books for UML is:
 - Martin Fowler, UML Distilled, 3rd Edition



UML diagram types

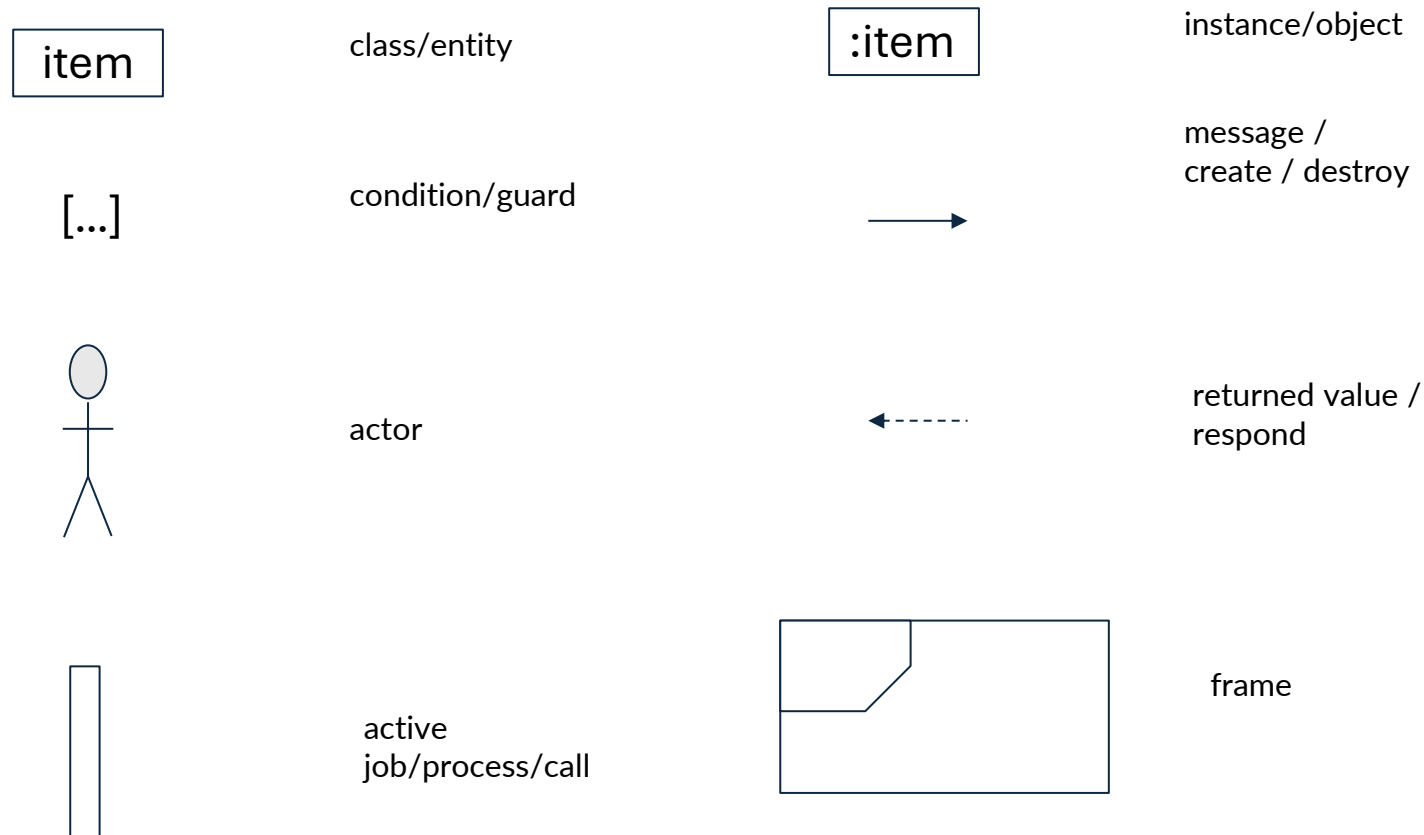
1. Activity diagrams: model activities involved in a process or in data processing
2. Use case diagrams: model interactions between a system and its environment. Used for requirements
3. Sequence diagrams: models interactions between people or sub-systems with other system components
4. Class diagrams: model object classes in the system and its properties including associations/relations between these classes
5. State diagrams: model how the system reacts to internal and external events

Interaction models

- Describe how group of objectives and components collaborate in some behavior
- Interactions can be
 - User interaction: inputs outputs
 - Component or systems interaction
- Modeling interaction of the system reveals communication problems between components
 - Good for identifying problems in requirements
- Help to understand the proposed system
- Examples of interaction models include
 - Use case modeling (previously given in requirement engineering lecture)
 - Sequence diagram

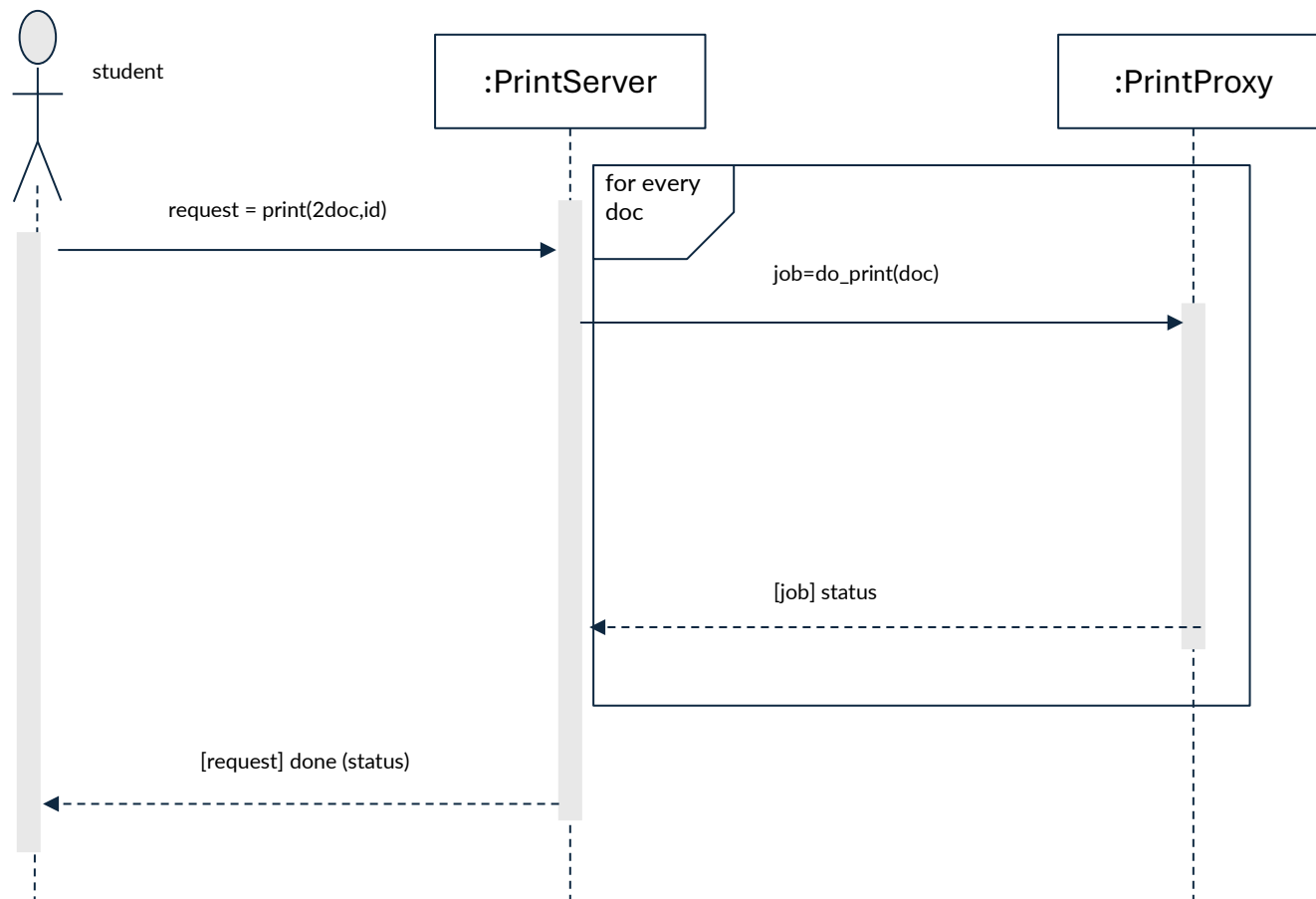
Sequence diagram

- Describes the behavior of a single scenario along with related group of objects
- X-axis is for objects and Y-axis is for time frame



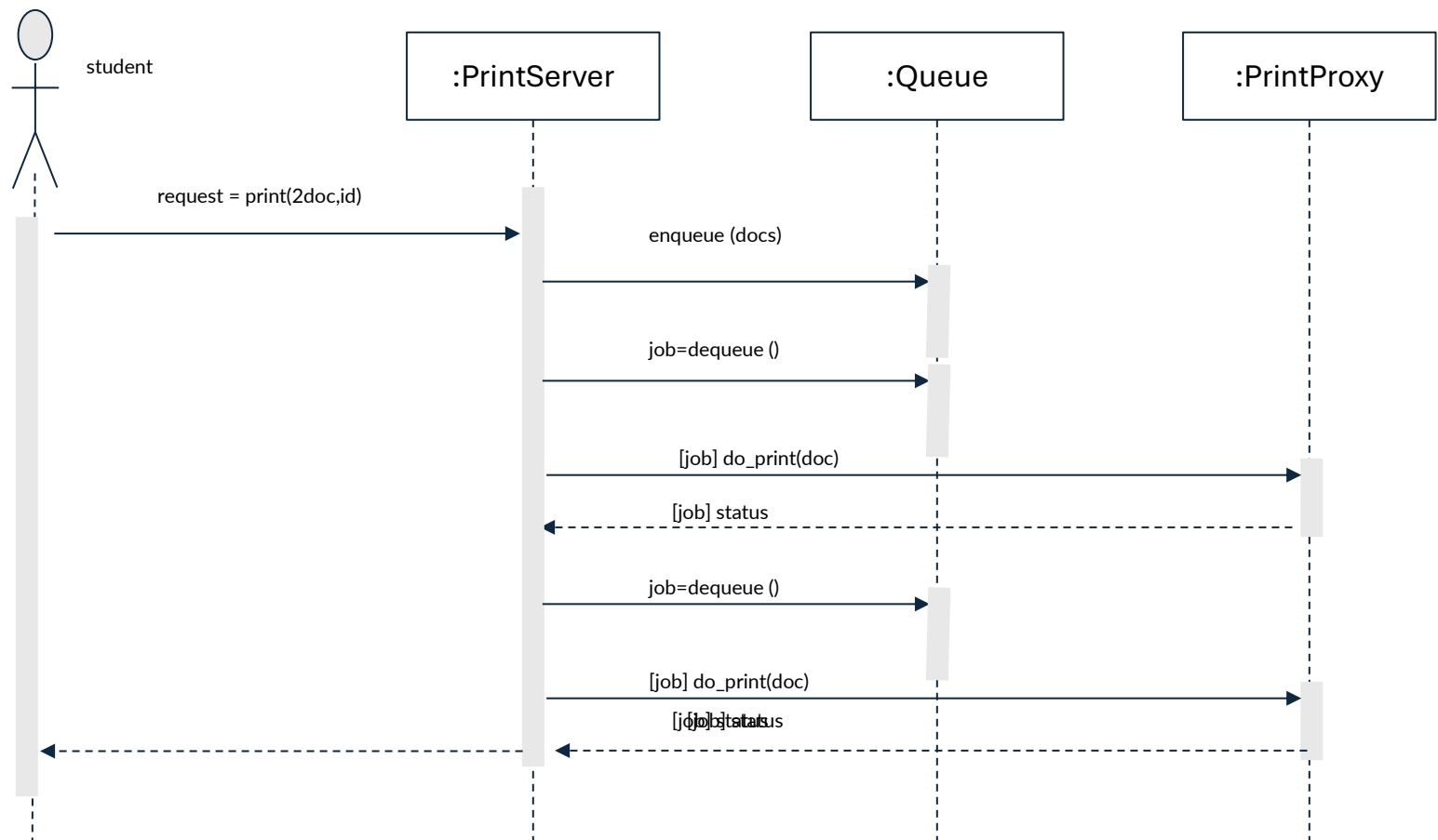
Sequence diagram (cont.)

- Example 1: student wants to print a couple of documents one at a time



Sequence diagram (cont.)

- Example 2: student wants to print a couple of documents simultaneously



Sequence diagram guidelines

- Identify the required scenario and related sub use cases
- Identify all interactive objects in the scenario: users, computers, etc
- Illustrate the scenario into a successful flow and alternate flow
 - Steps of scenario with interactive objects
- Identify the calling and responsive flow with objects

Sequence diagram example 1

A salesperson dispatches an order that may contain a number of items that are handled by different distributors. There are two types of distributors: critical one that handles items worth more than or equal to SR1000 and the other is a regular distributor that handles items less than that. Some customer may request confirmation of dispatch.

Sequence diagram example 2

Provide a sequence diagram that describes the actions that take place when a car is being booked by a customer. There are two types of booking: onsite using an automated machine or using the booking app. Regardless of booking method, the system must allow the booking only to customers who have less than 5 major traffic violations. The check must be made throughout the government query socket.

The objects that the sequence diagram must include are customer object, booking system, and government object.

Sequence diagram example 3

In the city library, a student can search and pick a book then go to the self-service loaning machine, which is connected to students' records. The student must pass the book QR code over the scanner of the machine. If the student has exceeded 3 books loans, then the system refuses the request. Otherwise, the book is lent to the student, and the library tracker is updated with the loaning information (student and book).